The Design of an 8 bit Split Capacitor Array DAC

Junming Zeng CID:01230747

***Abstract*—A chip design of an 180nm 8-bit split capacitor array digital to analogue converter (DAC) with serial input is presented in this paper. Starting from the clock generation until reaching the specified 0.4V to 1.4V analogue output range, a complete transistor level design is illustrated. With the 1.8V power supply and 10MHz clock speed, this DAC can achieve a maximum of 0.18% relative output error at the analogue output. The clock error is approximately 0.0006%, while the power is 5.68mW and 5.14mW for static and dynamic respectively. The total chip area is 0.04mm^2 while the main effort is to isolate the sensitive analogue part from the noisy digital part and to reduce the fabrication mismatches by applying common centroid.**

Keywords*—*Split Capacitor Array DAC, 8bit, 180nm, Layout.

I. INTRODUCTION

Mixed Signal Integrated Circuits usually comprise of a very important connection between the analogue components and the digital logics - Data Converters. Specifically, when doing digital signal processing designs, a Digital-to-Analogue-Converter (DAC) is in a high demand. For example in a Successive Approximate Register (SAR), the most critical component to design is normally the DAC[1], where the power consumption, Differential Nonlinearity (DNL), and Integral Nonlinearity (INL) dominant the performance[2]. This leads to different design topologies with different performance trade-offs. Table I shows the general comparison on the performance of the three selected DAC topologies.

TABLE I: DAC Topologies Comparison

Topology	Advantages	Disadvantages
R-2R	High Accuracy, Low Noise	High Glitch, Low Bandwidth[3]
Current Steering	High Speed, Low Glitch[4]	Matching Critical[5]
Capacitor Array	Simplicity, High Accuracy	Parasitic Capacitor, Matching[6]

Compared to current steering method, the most advantage of the capacitor array is its high accuracy, while on the contrary, low speed[7], and since in the design specification the output value will be sampled 1us after the start of conversion, a large tolerance in speed makes the design less critical to conversion time. Thus, the advantage of high speed in current steering can be neglected.

In addition, R-2R achieves high accuracy as well, and it can sufficiently occupy less chip area than the capacitor method[8]. However, its conversion time is the slowest among the three as every transition needs to redefine the current direction. Furthermore, random mismatch errors for resistors are considered as the most undesirable error in R-2R and it degrades the performance of the DAC[9].

Consequently, for this design, the capacitor array method is chosen and a further improvement of Split Capacitor Array is applied to reduce the total capacitance and therefore the chip area[10].

Section II will give an explanation on this mixed signal design. Section III will explain the important elements within the system especially the capacitor array. Section IV will evaluate the performance of the system. Finally, the conclusion will be made in Section V.

II. SYSTEM OVERVIEW

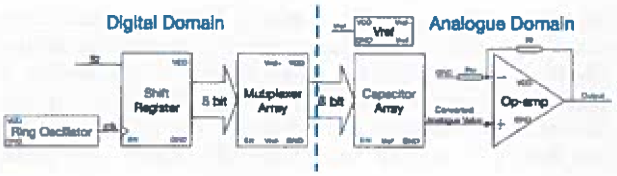


Fig. 1: The Top Level Schematic of DAC

Separated by the dashed line as shown in Figure 1, this system comprises of a digital domain and an analogue domain. At the digital part, a ring oscillator is used to generate the 10MHz clock and an 8-bit shift register is used to transfer the received serial data into parallel data out. The stages afterwards are all based on parallel processing of 8 bits. The multiplexer array is essentially the connection bridge between the digital and analogue blocks, where analogue reference voltages are selected and fed into the capacitor array. In the analogue part, the two reference voltages 0.4V and 1.4V are generated by the Vref block and this will be used as the lower and the upper limit of the analogue output. The capacitor array converts the low and high digital levels to the related analogue representations which is the main part of the DAC and the output is sent to a non-inverting amplifier to approach the desired reference.

III. CIRCUIT IMPLEMENTATION

In this section, the design of important blocks within the top level block diagram will be detailed explained, including the Ring Oscillator, Multiplexer, and the Capacitor Array.

A. Ring Oscillator

The Ring Oscillator is implemented using an odd number of inverters and an output buffer[11] as shown in Figure 2. The gate size of the inverter defines the resistance of it and therefore the delay time for a signal to reach from the input to the output. Odd number of inverters can effectively creates oscillation with a feedback loop and thus by changing the

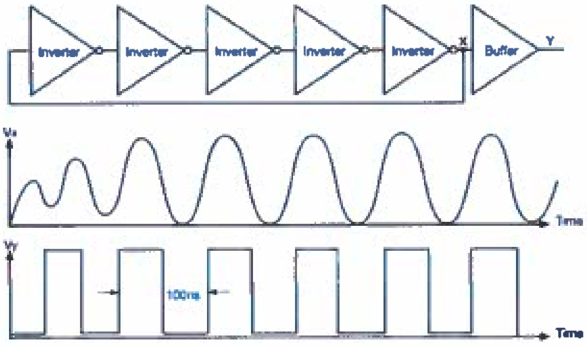


Fig. 2: The Schematic of Ring oscillator

delay of each inverter[12], a desired 10MHz oscillated signal can be generated. The output buffer is a specially designed inverter with a large gate width so that it can sharpen the output waveform to a square wave shaped clock (Y in Figure 2) instead of a sinc shaped wave (X). The rest of the system can benefit from a less rising time for a square wave clock.

$$f = \frac{1}{2 \times T \times n} \tag{1}$$

where f is the oscillation frequency, T is the time delay of a single inverter, and n is the number of inverter used.

B. Shift Register

The Shift Register is a Serial In Parallel Out (SIPO) structure[13]. The clock signal is gated (low power design) by the enable signal so that only when the enable signal is high, the flip-flops can then take samples at the rising edge of the clock.

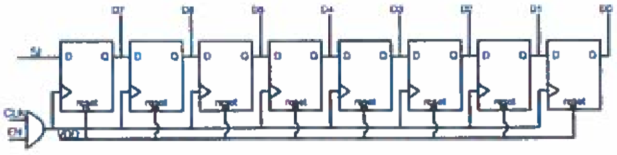


Fig. 3: The Schematic of Shift Register

C. Multiplexer Array

As the connection between the analogue and digital part, the Multiplexer Array consists a series of single pole double throw (SPDT) switches that each multiplexer selects the analogue reference voltage 0.4V (offset) and 1.4V (V_{ref+}) into the corresponding input port of the capacitor array. The inverter is controlling which voltage to select and the transmission gates support the voltage transmission, as shown in Figure 4.

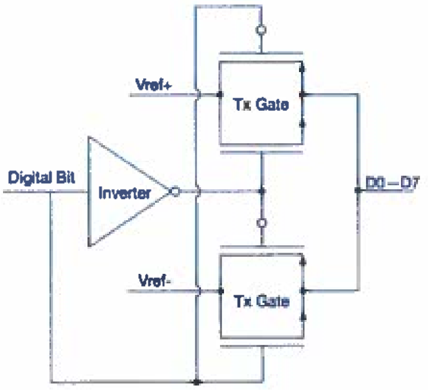


Fig. 4: The Schematic of 2-to-1 Multiplexer

D. Capacitor Array

The capacitor array is based on the model of charge distributed capacitor with binary weighted inputs[5]. It allows the voltage applied on the input of the array to be charge redistributed and output a voltage which is a portion defined by the input weight. A MOSFET switch is used to discharge the capacitor every time during the input being available at the shift register stage so that the array is reset. Then, after the serial data is converted to its parallel form and sent to the input of the capacitor array through the multiplexer switch, the switch opens so that the digital to analogue conversion begins.

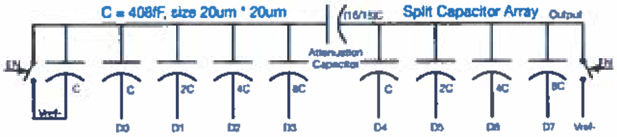


Fig. 5: The Schematic of Split Capacitor Array

An optimised method called the split capacitor array is applied with an attenuation capacitor connected in series between the four Least Significant Bits (LSB) and the four Most Significant Bits (MSB). whose value is calculated by[5]:

$$C_{atten} = \frac{\text{sum of the LSB array capacitors}}{\text{sum of the MSB array capacitors}} \times C = \frac{16}{15} C \tag{2}$$

The advantage of this method is it keeps the same weight of the input while reducing the overall size occupied.[14] By placing it at the middle of the string, the overall capacitance is at a minimum and this can be seen in Table II.

The value of the unit capacitor is 408fF with a size of 20um × 20um. As a result, the minimum attenuated capacitor value is equivalent to 400/16 = 25fF which is just larger than the unit parasitic capacitance in the layout (normally 10 to 20 fF). Besides, the input capacitance of the opamp is 300fF by simulation so the parasitic effect would ideally be less than 1 LSB.

TABLE II: Position of the Attenuation Capacitor with the Overall Capacitance Reduction

Position	Overall Capacitance	Reduction
No split	255C	0
2 bit LSB, 6 bit MSB	67.06C	74%
3 bit LSB, 5 bit MSB	39.25C	84.6%
4 bit LSB, 4 bit MSB	32.06C	87.4%

E. Rest of design

The reference voltage 0.4V and 1.4V are generated based on the provided 0.9V input voltage with active loads which provides an isolation from the digital power supply. The non-inverting amplifier at the last stage allows modifications at the final output so that the offset and gain error can be reduced accordingly. Due to the non-linearity and offset of the opamp at low input voltage, this DAC directly converts from 0.4V to 1.4V. Besides, the opamp also acts like a buffer to avoid the affect to the capacitor array.

IV. RESULTS

Overall, the result of ramping the serial input of the DAC from 00000000 to 11111111 (from the provided testbench) is shown in Figure 6, where the overall shape looks like a triangular. This is because every time when a new input is inserted, the capacitor array will be discharged by connecting the Vref- voltage directly across both sides of the capacitor array. Therefore, there is a reset followed by a charge up.

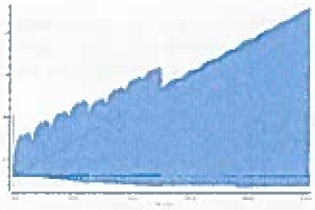
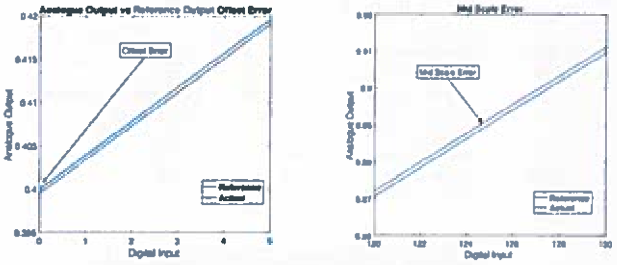


Fig. 6: The Overall Response by ramping the input

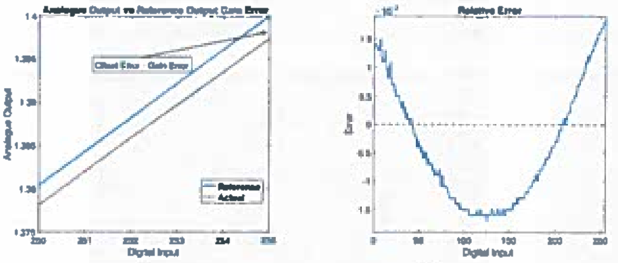
Generally, the actual DAC result is lower than the reference at the offset point, and grows slightly faster until approximately the mid scale. then it slows down the increment speed and reaches 0.18% maximum absolute relative error at the full scale point, which can be seen by the position of the two lines in Figure 7. This becomes clearer as plotted in Figure 7d where the error crosses the 0 error line twice which means it flipped from positive to negative and then back to positive. The maximum relative error is calculated by

$$ERR_{rel} = (VOUT_{ref} - VOUT_{out})/VOUT_{ref} \quad (3)$$

where in this design, modifications at the reference voltage and the gain of the opamp balances the maximum absolute error to be occurring at both the mid-scale and the full-scale, otherwise, a decrease in one will leads to an increase to the other.



(a) Offset Error (b) Mid Scale Error



(c) Gain Error (d) Relative Error

Fig. 7: Error results

The power simulation is realised by measuring the total current coming from the power supply and multiplying to the power supply voltage. This can be seen in Figure 8, where static power consumption mainly comes from the biasing of the circuit as shown at the bottom and the conversion contributes the dynamic power consumption. As the current level indicates, the static current is slightly higher than the dynamic current.

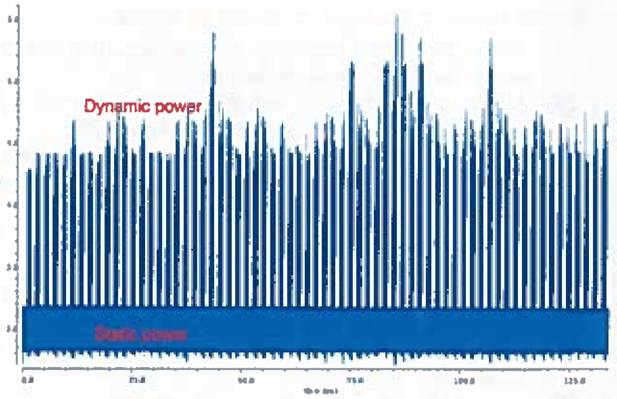


Fig. 8: The Power Supply Current

To summarise the total performance, the comparison between the specification and the actual result of the DAC is shown in Table III.

TABLE III: DAC Specification

Name	Specifications	Limit	Actual
Area	Total Silicon Area	$\leq 1mm^2$	$0.04mm^2$
P_{diss}	Power Dissipation	$\leq$ 20mW	5.68mW(S) + 5.14mW(D)
f_{clk}	Clock Frequency	10MHz	10MHz-0.0006%
f_s	Conversion Rate	500kHz	higher than 500kHz
R	Output Range	0.4-1.4V	0.4V-1.397V
ERR_{rel}	Relative Error	$\leq$ 1.6%	0.18%

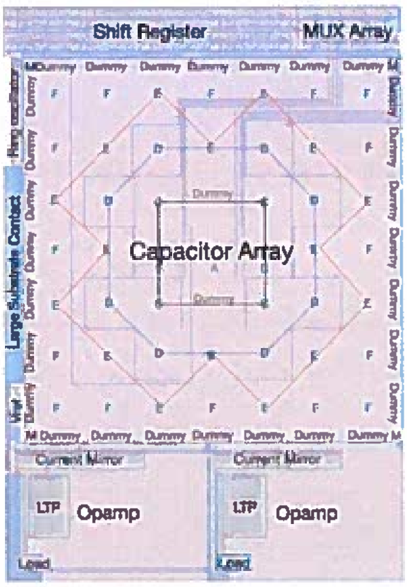


Fig. 9: The Layout of the DAC

In the layout of the DAC as shown in Figure 9,

1) The digital part is separated with the sensitive analogue part to avoid the noise influence.
2) All the current mirrors and differential pairs are made common centroid because they are sensitive.
3) Dummy polysilicon gates are added at every edge of the active area so that the source and drain are better aligned to reduce the edge effect.
4) Guard rings are added to provide isolations between the capacitor array, the opamp and the Ddigital part.
5) The capacitors are arranged with common centroid to reduce the mismatch, and dummy capacitors are added at the outer edge to reduce the edge mismatch effect[15].
6) The blank spaces are filled with substrate contacts to reduce the substrate to ground resistance.

V. CONCLUSION

This paper illustrated a design procedure of an 8-bit split capacitor array DAC from schematic to layout. The advantage of relatively good precision and simplicity of the capacitor array make it suitable to achieve the design requirements. There are a few works that has been explored but not included in the design: a further array splitting with two attenuation capacitors (but higher impact on parasitic capacitance), and a de-glitching capacitor array to reduce the dynamic power (but larger area). Generally, the effort is made mainly to find a design balance while it can still meets the specifications. Simulation results on the error and power consumption are investigated and the layout is demonstrated with the strategy used.

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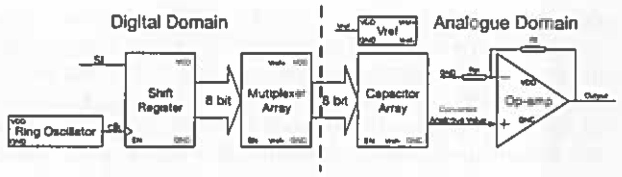

Fig. 1: The Top Level Schematic of DAC

Separated by the dashed line as shown in Figure 1, this system comprises of a digital domain and an analogue domain. At the digital part, a ring oscillator is used to generate the 10MHz clock and an 8-bit shift register is used to transfer the received serial data into parallel data out. The stages afterwards are all based on parallel processing of 8 bits. The multiplexer array is essentially the connection bridge between the digital and analogue blocks, where analogue reference voltages are selected and fed into the capacitor array. In the analogue part, the two reference voltages 0.4V and 1.4V are generated by the Vref block and this will be used as the lower and the upper limit of the analogue output. The capacitor array converts the low and high digital levels to the related analogue representations which is the main part of the DAC and the output is sent to a non-inverting amplifier to approach the desired reference.

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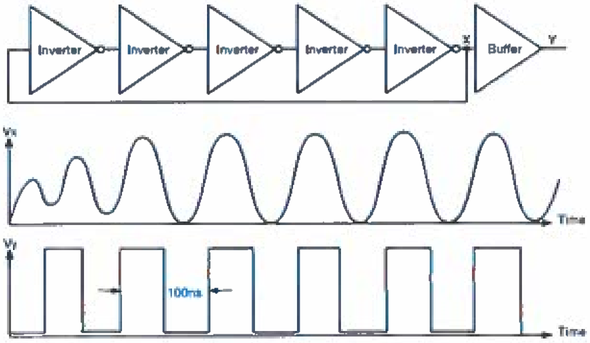

Fig. 2: The Schematic of Ring oscillator

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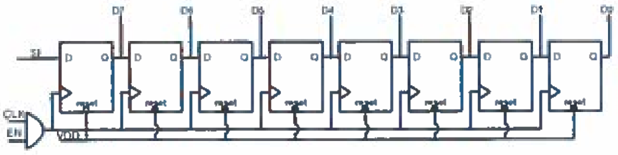

Fig. 3: The Schematic of Shift Register

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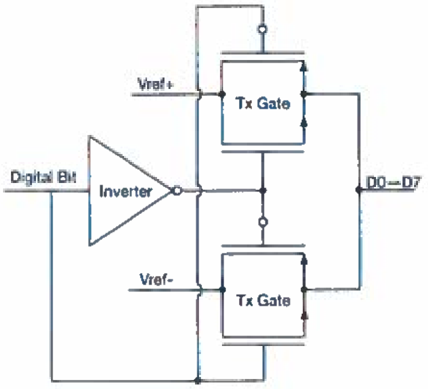

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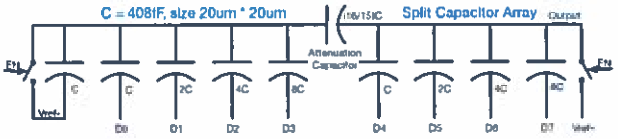

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Coursework Assessment

Full Custom Design of a Digital to Analogue Converter (DAC) in 0.18μm CMOS

jz11415

Jan, 2017

CONGRATULATIONS! Your DAC design is functional.

1 Overall Mark

Your Overall Design Mark is A
Your Overall Report Mark is A

2 Design Specifications

Table 1: Achieved Design Specifications

Name	Specifications	Value	Units	Rank	Best	Median
Area	Total silicon area	40800	μm^2	40% - 50%	6300.00	43056.00
P_{Sdiss}	Static Power dissipation	5.674	mW	40% - 50%	1.75	5.67
P_{Ddiss}	Dynamic Power dissipation	5.143	mW	40% - 50%	1.67	5.17
ERR_{rel}	Relative error	0.180	%	0 % - 10%	0.15	0.55
ERR_{clk}	Clock error	-0.010	%	10 % - 20%	0.00	0.42

Best is the best results across 41 students and Median is the median value

3 Comments On The Design

Please find comments for the design submission as below

3.1 Design and functionality

Binary C array based charge redistribution DAC. Unity cap array is used for splitter cap. Active potential devider based reference without buffering could have speed or non-linearity issue.

3.2 Design clarity

Clear top level design and sub cells, with necessary notes. Even custom made some symbols for illustrative purpose.

3.3 Layout

Your DAC layout passed DRC
Your DAC layout passed LVS

Layout in a rectangle shape. Dummy Res and resistor matching are not used. Dual row guarding contacts is used between cap array and ring osc. Can add more guard between cap array and digital.

4 Report Evaluation

Please find comments for your report as below

4.1 Design description

Good explanation of the design. The structure adopts first a high level description, then each block is detailed. You should have included the topology for your voltage reference generator, which should also be in a separate section. Schematics very clear, well done. You should have included a timing diagram.

4.2 Circuit Theory

Good background theory on the DAC topology. Some analysis is provided for the capacitor array and the ring oscillator. Table 1 is very good for design comparison, and your justification regarding the capacitor array is much appreciated. Some elements are unclear however, for instance the matching would be quite critical for an R/2R ladder as well, and I am not certain why this topology would be low noise. Overall, your circuit description might have included some more analysis on the transistor sizing and the design decisions.

4.3 Results and interpretation

Very interesting graphs, offset and gain errors are annotated on the figure. Power simulations are carried out, which is a very nice addition to your paper. As a note, keep in mind to consider the power through the 0.9 reference voltage as well, also it would have been interesting to look at the power consumption of each block separately, but these are just suggestions. The performance table is a good summary. Please provide a more accurate value for the sampling frequency.

4.4 Presentation

Good presentation of the paper, matches IEEE standards. The structure is clear, however some figures (especially Figs 6 and 8) need larger labels. Your abstract contains all the information, however make sure you start with a context sentence, follow with a summary of your design and end with the key results. The last sentence is not necessary except for the total chip area.

5 Simulation Log

Please find simulation log as below, please note these were the initial simulation results, and were reviewed by markers. Manual resimulated might be performed to get accurate specification for some points

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```
    This is evaluation results of Full Custom IC Design course work for jz11415
Creating test bench schematics (tcellDACforscript_jz11415 in CMOS16_DAC_jz11415) for design (DAC in CMOS16_DAC_jz11415)
Found terminal CLK of inst INST0
Matched to pin CLK in the assignment
Found terminal VOUT of inst INST0
Matched to pin VOUT in the assignment
Found terminal EN of inst INST0
Matched to pin EN in the assignment
Found terminal SI of inst INST0
Matched to pin SI in the assignment
Found terminal VREF of inst INST0
Matched to pin VREF in the assignment
CLK period is 99.9904 nS, relative error is -0.0096 Static power consumption is 5.6742 mW , dynamic power consumption is 5.1427 mW
Index Reference Vout error
0 0.4039 0.4034 0.0014
1 0.4078 0.4073 0.0014
2 0.4118 0.4112 0.0015
3 0.4157 0.4151 0.0014
4 0.4196 0.4191 0.0013
5 0.4235 0.4230 0.0013
6 0.4275 0.4269 0.0013
7 0.4314 0.4309 0.0012
8 0.4353 0.4348 0.0012
9 0.4392 0.4387 0.0012
10 0.4431 0.4426 0.0012
11 0.4471 0.4466 0.0011
12 0.4510 0.4504 0.0012
13 0.4549 0.4544 0.0011
14 0.4588 0.4584 0.0010
15 0.4627 0.4624 0.0008
16 0.4667 0.4663 0.0008
17 0.4706 0.4701 0.0009
18 0.4745 0.4741 0.0008
19 0.4784 0.4779 0.0010
```

```
20 0.4824 0.4820 0.0007
21 0.4863 0.4859 0.0007
22 0.4902 0.4899 0.0007
23 0.4941 0.4938 0.0007
24 0.4980 0.4978 0.0005
25 0.5020 0.5017 0.0006
26 0.5059 0.5056 0.0005
27 0.5098 0.5095 0.0005
28 0.5137 0.5135 0.0004
29 0.5176 0.5174 0.0005
30 0.5216 0.5214 0.0004
31 0.5255 0.5252 0.0005
32 0.5294 0.5293 0.0002
33 0.5333 0.5331 0.0004
34 0.5373 0.5372 0.0001
35 0.5412 0.5411 0.0002
36 0.5451 0.5451 0.0000
37 0.5490 0.5490 0.0001
38 0.5529 0.5529 0.0000
39 0.5569 0.5568 0.0001
40 0.5608 0.5608 0.0001
41 0.5647 0.5647 0.0000
42 0.5686 0.5687 0.0001
43 0.5725 0.5726 0.0001
44 0.5765 0.5766 0.0002
45 0.5804 0.5805 0.0002
46 0.5843 0.5844 0.0002
47 0.5882 0.5885 0.0004
48 0.5922 0.5923 0.0003
49 0.5961 0.5963 0.0003
50 0.6000 0.6002 0.0003
51 0.6039 0.6042 0.0005
52 0.6078 0.6081 0.0004
53 0.6118 0.6120 0.0005
54 0.6157 0.6160 0.0005
55 0.6196 0.6199 0.0005
56 0.6235 0.6239 0.0006
57 0.6275 0.6278 0.0006
58 0.6314 0.6318 0.0007
59 0.6353 0.6357 0.0007
60 0.6392 0.6397 0.0007
61 0.6431 0.6436 0.0007
```

```
62 0.6471 0.6475 0.0007
63 0.6510 0.6515 0.0007
64 0.6549 0.6553 0.0007
65 0.6588 0.6593 0.0008
66 0.6627 0.6633 0.0008
67 0.6667 0.6672 0.0009
68 0.6706 0.6712 0.0009
69 0.6745 0.6751 0.0008
70 0.6784 0.6790 0.0009
71 0.6824 0.6830 0.0009
72 0.6863 0.6870 0.0010
73 0.6902 0.6909 0.0010
74 0.6941 0.6948 0.0010
75 0.6980 0.6987 0.0009
76 0.7020 0.7027 0.0011
77 0.7059 0.7066 0.0010
78 0.7098 0.7106 0.0011
79 0.7137 0.7145 0.0011
80 0.7176 0.7185 0.0012
81 0.7216 0.7224 0.0011
82 0.7255 0.7264 0.0013
83 0.7294 0.7303 0.0013
84 0.7333 0.7342 0.0012
85 0.7373 0.7382 0.0013
86 0.7412 0.7422 0.0013
87 0.7451 0.7460 0.0013
88 0.7490 0.7500 0.0013
89 0.7529 0.7539 0.0013
90 0.7569 0.7579 0.0014
91 0.7608 0.7618 0.0014
92 0.7647 0.7657 0.0013
93 0.7686 0.7697 0.0014
94 0.7725 0.7735 0.0013
95 0.7765 0.7776 0.0014
96 0.7804 0.7815 0.0014
97 0.7843 0.7854 0.0014
98 0.7882 0.7894 0.0015
99 0.7922 0.7933 0.0014
100 0.7961 0.7973 0.0015
101 0.8000 0.8012 0.0015
102 0.8039 0.8052 0.0015
103 0.8078 0.8091 0.0015
```

104	0.8118	0.8130	0.0016
105	0.8157	0.8169	0.0015
106	0.8196	0.8209	0.0016
107	0.8235	0.8248	0.0016
108	0.8275	0.8288	0.0016
109	0.8314	0.8327	0.0016
110	0.8353	0.8366	0.0016
111	0.8392	0.8405	0.0016
112	0.8431	0.8445	0.0016
113	0.8471	0.8484	0.0016
114	0.8510	0.8523	0.0016
115	0.8549	0.8563	0.0016
116	0.8588	0.8602	0.0016
117	0.8627	0.8642	0.0017
118	0.8667	0.8680	0.0016
119	0.8706	0.8720	0.0016
120	0.8745	0.8759	0.0016
121	0.8784	0.8799	0.0016
122	0.8824	0.8837	0.0016
123	0.8863	0.8877	0.0016
124	0.8902	0.8916	0.0016
125	0.8941	0.8956	0.0016
126	0.8980	0.8995	0.0016
127	0.9020	0.9034	0.0016
128	0.9059	0.9073	0.0016
129	0.9098	0.9113	0.0016
130	0.9137	0.9152	0.0016
131	0.9176	0.9191	0.0016
132	0.9216	0.9231	0.0016
133	0.9255	0.9269	0.0016
134	0.9294	0.9309	0.0016
135	0.9333	0.9348	0.0015
136	0.9373	0.9387	0.0016
137	0.9412	0.9426	0.0016
138	0.9451	0.9466	0.0016
139	0.9490	0.9505	0.0015
140	0.9529	0.9544	0.0016
141	0.9569	0.9583	0.0015
142	0.9608	0.9623	0.0016
143	0.9647	0.9662	0.0015
144	0.9686	0.9701	0.0016
145	0.9725	0.9741	0.0016

```
146 0.9765 0.9780 0.0016
147 0.9804 0.9819 0.0016
148 0.9843 0.9858 0.0016
149 0.9882 0.9897 0.0015
150 0.9922 0.9936 0.0014
151 0.9961 0.9975 0.0014
152 1.0000 1.0015 0.0015
153 1.0039 1.0053 0.0014
154 1.0078 1.0093 0.0014
155 1.0118 1.0132 0.0014
156 1.0157 1.0171 0.0014
157 1.0196 1.0210 0.0014
158 1.0235 1.0249 0.0013
159 1.0275 1.0288 0.0013
160 1.0314 1.0328 0.0014
161 1.0353 1.0367 0.0014
162 1.0392 1.0405 0.0013
163 1.0431 1.0445 0.0013
164 1.0471 1.0484 0.0013
165 1.0510 1.0523 0.0013
166 1.0549 1.0562 0.0013
167 1.0588 1.0601 0.0012
168 1.0627 1.0640 0.0012
169 1.0667 1.0679 0.0011
170 1.0706 1.0718 0.0012
171 1.0745 1.0757 0.0011
172 1.0784 1.0796 0.0011
173 1.0824 1.0835 0.0011
174 1.0863 1.0875 0.0011
175 1.0902 1.0913 0.0010
176 1.0941 1.0953 0.0010
177 1.0980 1.0992 0.0010
178 1.1020 1.1030 0.0009
179 1.1059 1.1069 0.0009
180 1.1098 1.1109 0.0010
181 1.1137 1.1147 0.0009
182 1.1176 1.1185 0.0008
183 1.1216 1.1225 0.0008
184 1.1255 1.1263 0.0007
185 1.1294 1.1303 0.0008
186 1.1333 1.1341 0.0007
187 1.1373 1.1381 0.0007
```

```
188 1.1412 1.1420 0.0007
189 1.1451 1.1459 0.0007
190 1.1490 1.1497 0.0006
191 1.1529 1.1536 0.0006
192 1.1569 1.1575 0.0005
193 1.1608 1.1614 0.0005
194 1.1647 1.1653 0.0005
195 1.1686 1.1691 0.0004
196 1.1725 1.1731 0.0005
197 1.1765 1.1770 0.0004
198 1.1804 1.1809 0.0004
199 1.1843 1.1847 0.0003
200 1.1882 1.1886 0.0003
201 1.1922 1.1925 0.0003
202 1.1961 1.1964 0.0003
203 1.2000 1.2002 0.0002
204 1.2039 1.2041 0.0002
205 1.2078 1.2080 0.0001
206 1.2118 1.2118 0.0001
207 1.2157 1.2158 0.0001
208 1.2196 1.2196 0.0000
209 1.2235 1.2235 0.0000
210 1.2275 1.2274 0.0000
211 1.2314 1.2313 0.0001
212 1.2353 1.2351 0.0001
213 1.2392 1.2389 0.0002
214 1.2431 1.2429 0.0002
215 1.2471 1.2467 0.0003
216 1.2510 1.2506 0.0003
217 1.2549 1.2544 0.0004
218 1.2588 1.2584 0.0004
219 1.2627 1.2622 0.0004
220 1.2667 1.2661 0.0005
221 1.2706 1.2700 0.0005
222 1.2745 1.2739 0.0005
223 1.2784 1.2777 0.0006
224 1.2824 1.2815 0.0007
225 1.2863 1.2854 0.0007
226 1.2902 1.2893 0.0007
227 1.2941 1.2931 0.0008
228 1.2980 1.2970 0.0008
229 1.3020 1.3008 0.0009
```

```
230 1.3059 1.3047 0.0009
231 1.3098 1.3086 0.0009
232 1.3137 1.3124 0.0010
233 1.3176 1.3162 0.0011
234 1.3216 1.3201 0.0011
235 1.3255 1.3240 0.0011
236 1.3294 1.3279 0.0012
237 1.3333 1.3317 0.0012
238 1.3373 1.3356 0.0012
239 1.3412 1.3395 0.0013
240 1.3451 1.3433 0.0013
241 1.3490 1.3472 0.0013
242 1.3529 1.3511 0.0014
243 1.3569 1.3549 0.0015
244 1.3608 1.3588 0.0014
245 1.3647 1.3626 0.0015
246 1.3686 1.3665 0.0016
247 1.3725 1.3704 0.0016
248 1.3765 1.3742 0.0016
249 1.3804 1.3781 0.0017
250 1.3843 1.3820 0.0017
251 1.3882 1.3858 0.0017
252 1.3922 1.3897 0.0018
253 1.3961 1.3936 0.0018
254 1.4000 1.3974 0.0018
Max ERROR is 0.0018
Simulation finished
```

Analogue Design Logbook for Lab2

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CID:01230747

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1 Inverter

To help building a layout diagram, a stick diagram is firstly produced and it is generated based on the schematic diagram in Lab 1. According to this stick diagram, the layout can be seen in Figure 1. This inverter is formed of a PMOS at the top with an n well underneath and an NMOS at the bottom. The total size is $2.96\mu m \times 8\mu m$ where the height is limited by the requirement of the standard cell and the width is dependent of the design rule of the gpdk180 library. Two fingers are used instead of one at the layout of the PMOS in order to reduce the height for individual finger. Therefore, it leaves a relatively large area for metal 1 routing in the later stages between different cells.

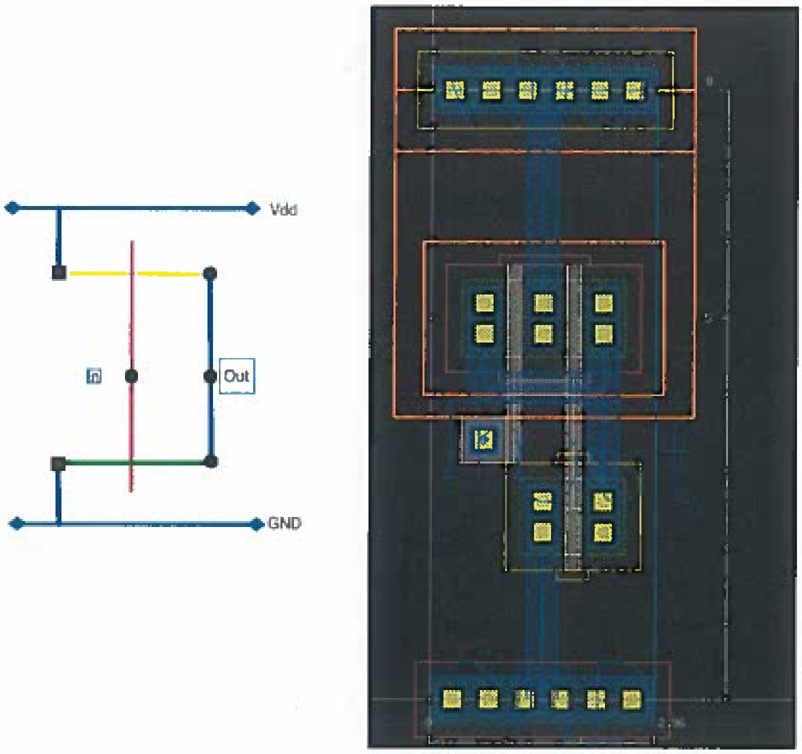
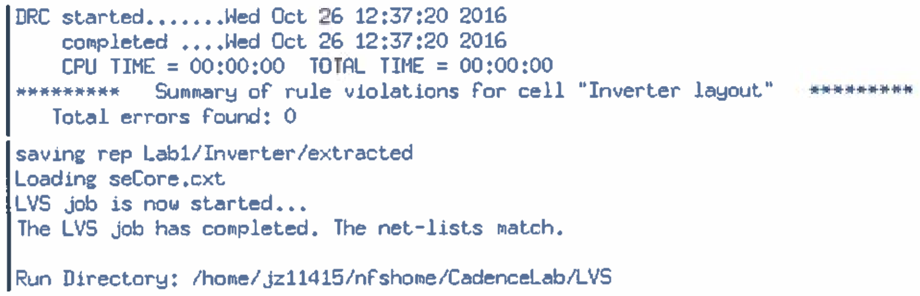


Figure 1: The Layout Diagram of the Inverter



```
DRC started.......Wed Oct 26 12:37:20 2016
    completed ....Wed Oct 26 12:37:20 2016
    CPU TIME = 00:00:00  TOTAL TIME = 00:00:00
*********   Summary of rule violations for cell "Inverter layout"   *********
   Total errors found: 0
saving rep Lab1/Inverter/extracted
Loading seCore.cxt
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/jz11415/nfshome/CadenceLab/LVS
```

Figure 2: The DRC and LVS check of the Inverter

2 NAND Gate

The NAND gate follows the function that $Output = \overline{A \cdot B}$ and thus $\overline{Output} = A \cdot B$ and $Output = \bar{A} + \bar{B}$. Intuitively, the NMOS should be formed that when both inputs A and B are high, the output should be shorted to ground via NMOS (connected in series). Similarly, the PMOS should be formed that either A or B is low, PMOS should conduct the V_{dd} to the V_{out} (connected in parallel). The schematic can be shown in Figure 3.

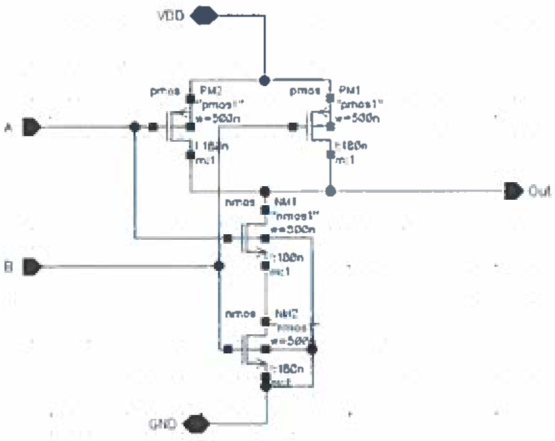


Figure 3: The Schematic Diagram of the NAND Gate

There are two design criteria:

1. The switching point should be half way on the V_{dd} and ground and therefore $0.9V$.
2. The propogation delay with a single pin changes should not exceed $10ns$.

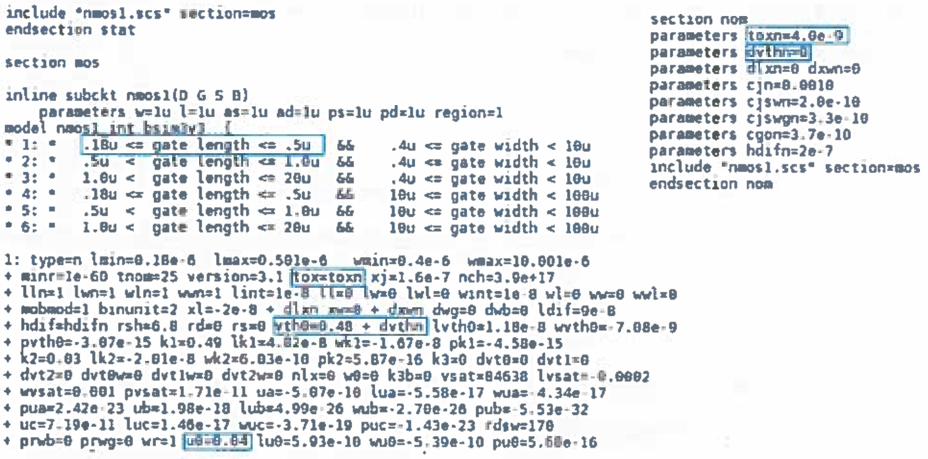
Based on the siwtching point equation from [1], where μ is the mobility, ϵ is the permittivity and t_{ox} is the thickness of the oxide.

$$V_{sp} = \frac{\sqrt{\frac{\beta_n}{N^2\beta_p}} \times V_{THN} + (V_{DD} - V_{THP})}{1 + \sqrt{\frac{\beta_n}{N^2\beta_p}}} \tag{1}$$

$$\text{where } \beta_n = \frac{\mu_{0n}\epsilon_r\epsilon_0}{t_{oxn}},$$
$$\beta_p = \frac{\mu_{0p}\epsilon_r\epsilon_0}{t_{oxp}}$$

Since we already have $V_{sp} = 0.9$, N = 2 (2 input NAND) and $V_{DD} = 1.8$. The variables that needs to be found out are β_n, β_p, V_{THN}, and V_{THP}.

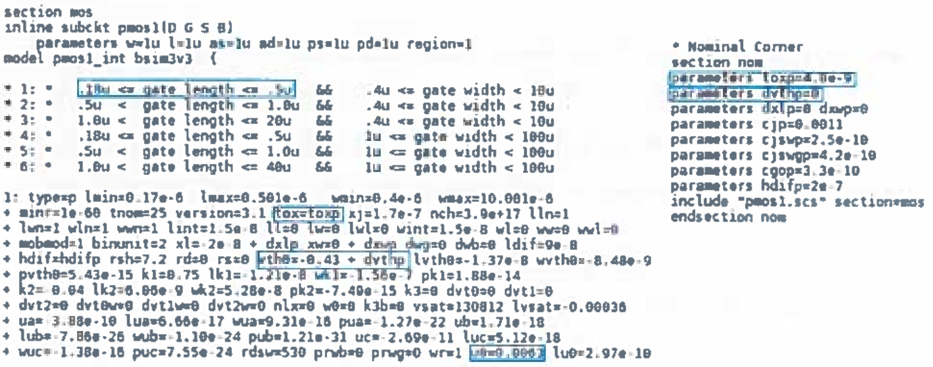
These can be found in the device property files located at /usr/local/cadence/kits/gpdk/180/models/spectre/, in which more specifically nmos1.scs and pmos1.scs. The screenshot of the key parameters has been highlighted in Figure 4 (NMOS), and Figure 5 (PMOS).



```
include "nmos1.scs" section=mos
endsection stat

section mos

inline subckt nmos1(D G S B)
    parameters w=1u l=1u as=1u ad=1u ps=1u pd=1u region=1
model nmos1_int bsim3v3  {
* 1: *    .18u <= gate length <= .5u   &&    .4u <= gate width < 10u
* 2: *    .5u  <  gate length <= 1.0u  &&    .4u <= gate width < 10u
* 3: *    1.0u <  gate length <= 20u   &&    .4u <= gate width < 10u
* 4: *    .18u <= gate length <= .5u   &&    10u <= gate width < 100u
* 5: *    .5u  <  gate length <= 1.0u  &&    10u <= gate width < 100u
* 6: *    1.0u <  gate length <= 20u   &&    10u <= gate width < 100u

1: type=n lmin=0.18e-6  lmax=0.501e-6   wmin=0.4e-6  wmax=10.001e-6
+ minr=1e-60 tnom=25 version=3.1 tox=toxn xj=1.6e-7 nch=3.9e+17
+ lln=1 lwn=1 wln=1 wwn=1 lint=1e-8 ll=0 lw=0 lwl=0 wint=1e-8 wl=0 ww=0 wwl=0
+ mobmod=1 binunit=2 xl=-2e-8 + dlxn xw=0 + dxwn dwg=0 dwb=0 ldif=9e-8
+ hdif=hdifn rsh=6.8 rd=0 rs=0 vth0=0.48 + dvthn lvth0=1.18e-8 wvth0=-7.08e-9
+ pvth0=-3.07e-15 k1=0.49 lk1=4.82e-8 wk1=-1.67e-8 pk1=-4.58e-15
+ k2=0.03 lk2=-2.01e-8 wk2=6.03e-10 pk2=5.87e-16 k3=0 dvt0=0 dvt1=0
+ dvt2=0 dvt0w=0 dvt1w=0 dvt2w=0 nlx=0 w0=0 k3b=0 vsat=84638 lvsat=-0.0002
+ wvsat=0.001 pvsat=1.71e-11 ua=-5.07e-10 lua=-5.58e-17 wua=-4.34e-17
+ pua=2.42e-23 ub=1.98e-18 lub=4.99e-26 wub=-2.70e-26 pub=-5.53e-32
+ uc=7.19e-11 luc=1.46e-17 wuc=-3.71e-19 puc=-1.43e-23 rdsw=170
+ prwb=0 prwg=0 wr=1 u0=0.04 lu0=5.93e-10 wu0=-5.39e-10 pu0=5.60e-16
```

```
section nom
parameters toxn=4.0e-9
parameters dvthn=0
parameters dlxn=0 dxwn=0
parameters cjn=0.0010
parameters cjswn=2.0e-10
parameters cjswgn=3.3e-10
parameters cgon=3.7e-10
parameters hdifn=2e-7
include "nmos1.scs" section=mos
endsection nom
```

Figure 4: The Parameter of NMOS in gpdk180



```
section mos
inline subckt pmos1(D G S B)
    parameters w=1u l=1u as=1u ad=1u ps=1u pd=1u region=1
model pmos1_int bsim3v3  {

* 1: *    .18u <= gate length <= .5u   &&    .4u <= gate width < 10u
* 2: *    .5u  <  gate length <= 1.0u  &&    .4u <= gate width < 10u
* 3: *    1.0u <  gate length <= 20u   &&    .4u <= gate width < 10u
* 4: *    .18u <= gate length <= .5u   &&    1u <= gate width < 100u
* 5: *    .5u  <  gate length <= 1.0u  &&    1u <= gate width < 100u
* 6: *    1.0u <  gate length <= 40u   &&    1u <= gate width < 100u

1: type=p lmin=0.17e-6  lmax=0.501e-6   wmin=0.4e-6  wmax=10.001e-6
+ minr=1e-60 tnom=25 version=3.1 tox=toxp xj=1.7e-7 nch=3.9e+17 lln=1
+ lwn=1 wln=1 wwn=1 lint=1.5e-8 ll=0 lw=0 lwl=0 wint=1.5e-8 wl=0 ww=0 wwl=0
+ mobmod=1 binunit=2 xl=-2e-8 + dxlp xw=0 + dxwp dwg=0 dwb=0 ldif=9e-8
+ hdif=hdifp rsh=7.2 rd=0 rs=0 vth0=-0.43 + dvthp lvth0=-1.37e-8 wvth0=-8.48e-9
+ pvth0=5.43e-15 k1=0.75 lk1=-1.21e-8 wk1=-1.56e-7 pk1=1.88e-14
+ k2=-0.04 lk2=6.06e-9 wk2=5.28e-8 pk2=-7.40e-15 k3=0 dvt0=0 dvt1=0
+ dvt2=0 dvt0w=0 dvt1w=0 dvt2w=0 nlx=0 w0=0 k3b=0 vsat=130812 lvsat=-0.00036
+ ua=-3.88e-10 lua=6.66e-17 wua=9.31e-18 pua=-1.27e-22 ub=1.71e-18
+ lub=-7.86e-26 wub=-1.10e-24 pub=1.21e-31 uc=-2.69e-11 luc=5.12e-18
+ wuc=-1.38e-16 puc=7.55e-24 rdsw=530 prwb=0 prwg=0 wr=1 u0=0.0063 lu0=2.97e-10
```

```
* Nominal Corner
section nom
parameters toxp=4.0e-9
parameters dvthp=0
parameters dxlp=0 dxwp=0
parameters cjp=0.0011
parameters cjswp=2.5e-10
parameters cjswgp=4.2e-10
parameters cgop=3.3e-10
parameters hdifp=2e-7
include "pmos1.scs" section=mos
endsection nom
```

Figure 5: The Parameter of PMOS in gpdk180

This gives, for NMOS $t_{ox} = t_{oxn} = 4.0e^{-9}, dvthn = 0, vtho = 0.48 + dvthn = 0.48, \mu_{0n} = 0.04$, and for PMOS $t_{ox} = t_{oxn} = 4.0e^{-9}, dvthn = 0, vtho = -0.43 + dvthn = 0.48, \mu_{0p} = 0.0063$.

Assuming the gate length of both PMOS transistors and NMOS transistors are the same, together with the same thickness of the oxide as well as the permittivity. The equation 1 can be simplified into:

$$V_{sp} = \frac{\sqrt{\frac{0.04\,W_n}{2^2 \cdot 0.0063\,W_p}} \times 0.48 + 1.8 - 0.43}{1 + \sqrt{\frac{0.04 \cdot W_n}{2^2 \cdot 0.0063 \cdot W_p}}} = 0.9V \tag{2}$$

$$\frac{W_n}{W_p} \approx 0.8$$

By calculating the switching point, we can get the approximate ratio of the PMOS width and NMOS width. In addition, the ratio of the width over the length of a single transistor also needs to be calculated. This should be done by the limitation of the delay time.

To achieve the delay less than 10ns as the design requirement states, the first thing to notice is the relationship of W/L and the propagation delay from high to low (PHL) and from low to high(PLH), which is given in Equation 3 and Equation 4 [1] with the assumption that the load capacitance is much larger than the total output capacitance from the MOS transistors.

$$T_{PHL} = 0.7 \times R_n \times N \times C_{load} \tag{3}$$

$$T_{PLH} = 0.7 \times \frac{R_p}{N} \times C_{load} \tag{4}$$

Since N and C_{load} are fixed. T_{PHL} is directly proportional to R (output equivalent resistance of the MOSFETs) and $R \approx \frac{1}{\lambda I_d}$, where I_d is given by

$$I_d = \frac{\mu \epsilon W}{L}((V_{GS} - V_t)V_{DS} - \frac{V_{DS}^2}{2})(1 + \lambda V_{DS}) \tag{5}$$

As seen in the equation, T_{PHL} and T_{PLH} are inversely proportional to W/L.The length has already been fixed to 180nm so that the minimum area can be achieved, the larger the gate width, the smaller the propagation delay is. However, the larger the transistor and therefore the larger the total area is going to be. According to the design requirement, as long as the delay is shorter than 10ns it is acceptable. So this leads to the next step.

Calculate the minimum gate width so that the propagation delay is less than 10ns. As shown in the table from the lecture notes, this 180nm (long channel) gate has an approximate resistance of $R_n = 15kW/L$ and $R_p = 45kW/L$.

$$T_{PHL} = 0.7 \times R_n \times N \times C_{load} < 10ns$$
$$T_{PLH} = 0.7 \times \frac{R_p}{N} \times C_{load} < 10ns$$

where $C_{load} = 1pF$ (load capacitance) and $N = 2$ (2 input NAND gate).

These gives $W_n > 378nm$ and $W_p > 283nm$ representing the width of NMOS and PMOS respectively.

For this 180nm technology, the minimum gate width is 400nm. To leave some tolerance of the modification when simulation, the NMOS gate is chosen by 500nm and the PMOS gate width is chosen by $500nm \times 0.8 = 400nm$, which both of the parameters satisfy the above minimum width condition.

To simulate the switching point of the NAND gate, Two inputs of the NAND gates are tied together so that the input will change from A=0 B=0 to A=1 B=1. In theory, the output should swing from 1.8V to 0V. By applying the parameter from the calculation, $W_n = 500nm$ and $W_p = 400nm$, it does not give a desired switching point. In this scenario, the switching point is closer to the 0V level as seen in the very left red line in Figure 6. To balance this, the PMOS needs to grow 'stronger' by increasing the width so that the switching point would be 'dragged' towards the middle point(to the left). This has been done by setting the width of the PMOS to a parameter w and sweep the parameter from 400nm to 600nm to find an approximate value and in addition, sweep with smaller unit to get a more accurate result. The result can be shown in Figure 7 where 500nm gives the most accurate result(highlighted) compares to the rest. Although the switching point can actually be swept to the exact middle point of the voltage swing which is 0.9V, the minimum unit that can be realised onto the gate is 5nm which limits the accuracy. The final switching point is fixed at 899.6mV with the width of PMOS is the same as the width of NMOS, i.e. $\frac{W_n}{W_p} = 1$

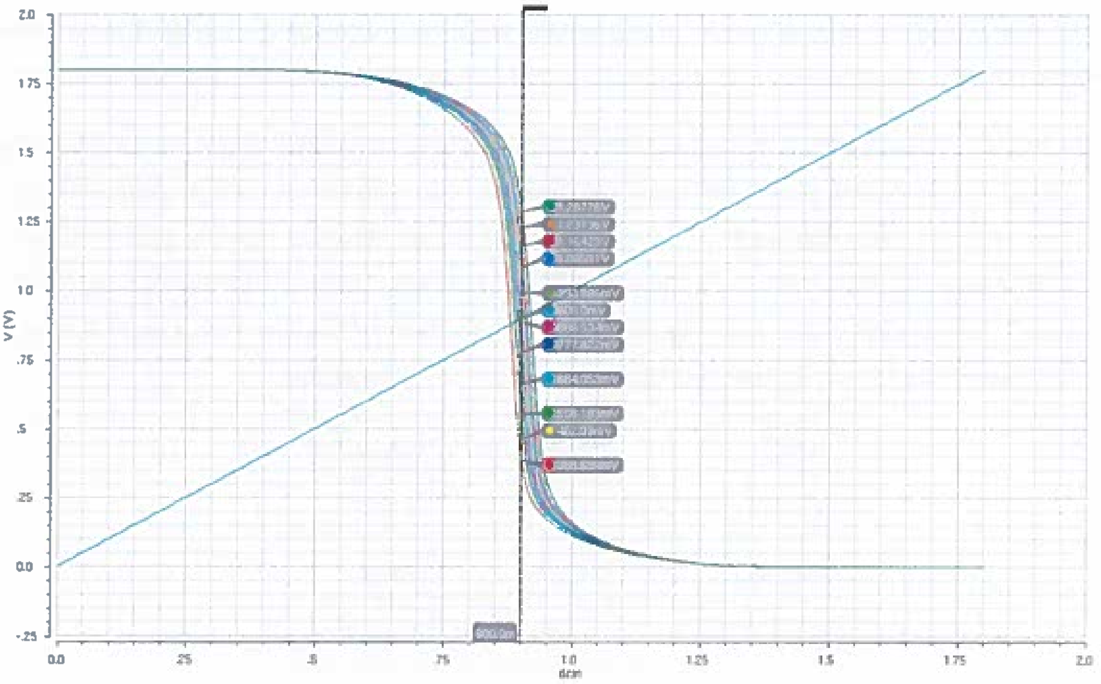
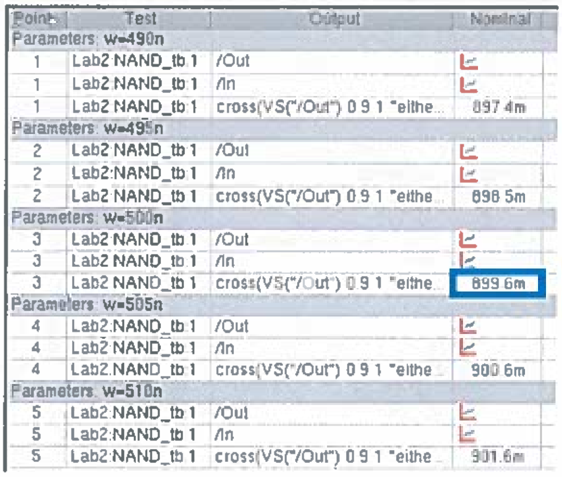


Figure 6: Sweeping the Width of the PMOS



Point	Test	Output	Nominal
Parameters: w=490n			
1	Lab2:NAND_tb:1	/Out	
1	Lab2:NAND_tb:1	/In	
1	Lab2:NAND_tb:1	cross(VS("/Out") 0.9 1 "eithe...	897.4m
Parameters: w=495n			
2	Lab2:NAND_tb:1	/Out	
2	Lab2:NAND_tb:1	/In	
2	Lab2:NAND_tb:1	cross(VS("/Out") 0.9 1 "eithe...	898.5m
Parameters: w=500n			
3	Lab2:NAND_tb:1	/Out	
3	Lab2:NAND_tb:1	/In	
3	Lab2:NAND_tb:1	cross(VS("/Out") 0.9 1 "eithe...	899.6m
Parameters: w=505n			
4	Lab2:NAND_tb:1	/Out	
4	Lab2:NAND_tb:1	/In	
4	Lab2:NAND_tb:1	cross(VS("/Out") 0.9 1 "eithe...	900.6m
Parameters: w=510n			
5	Lab2:NAND_tb:1	/Out	
5	Lab2:NAND_tb:1	/In	
5	Lab2:NAND_tb:1	cross(VS("/Out") 0.9 1 "eithe...	901.6m

Figure 7: Switching Point by Sweeping the Width of the PMOS

The propagation delay is simulated by fixing one input high and switching the other one. This is because, in the NAND gate, the worst case of propagation delay happens when only one of the PMOS starts conducting. Since two PMOS are connecting in parallel between the VDD and the output, the equivalent resistance of them two is $R_{eq} = R_{PMOS1}||R_{PMOS2}$. Moreover, the total propagation delay is proportional to the resistance. As a result, if the two input are tied together to measure the propagation delay, this would cause the PMOS to be switched on together and therefore the active resistance is half of only switching on one PMOS resulting in the propagation delay time is half as the worst case. So the worst case need to be found out by fixing one input high(cut off one PMOS) and switching on the other PMOS.

In theory, T_{PHL} is half of T_{PLH} given in this test all the MOSFETs have the same width and length. This is because the charge taken to rise up the output voltage needs to go through only one PMOS(relative resistance R) but the charge taken to drag down the output voltage needs to go through both NMOS in series(relative resistance 2R).

Table 1 shows the propagation delay simulation results of three cases, either tied both input together and switching them, or fixed A or fixed B. As it shows, the later two cases gives the same T_{PHL} as the first case while they also give twice the delay time on T_{PLH}. There are no significant difference between fixed A switch B and fixed B switch A because all the transistors share the same parameters. However, the tiny difference still states that by fixing B and switching A, it gives the worst propagation delay of $T_{PLH} = 7.92ns$ and $T_{PHL} = 4.23ns$ as shown in Figure 8 which both of them satisfy the desired requirement of 10ns.

Cases	T_{PHL} (ns)	$T_{PLH}(ns)$
Switching A and B	4.234	4.037
Switching A	4.231	7.921
Switching B	4.227	7.918

Table 1: Propagation Comparison for the Three Cases

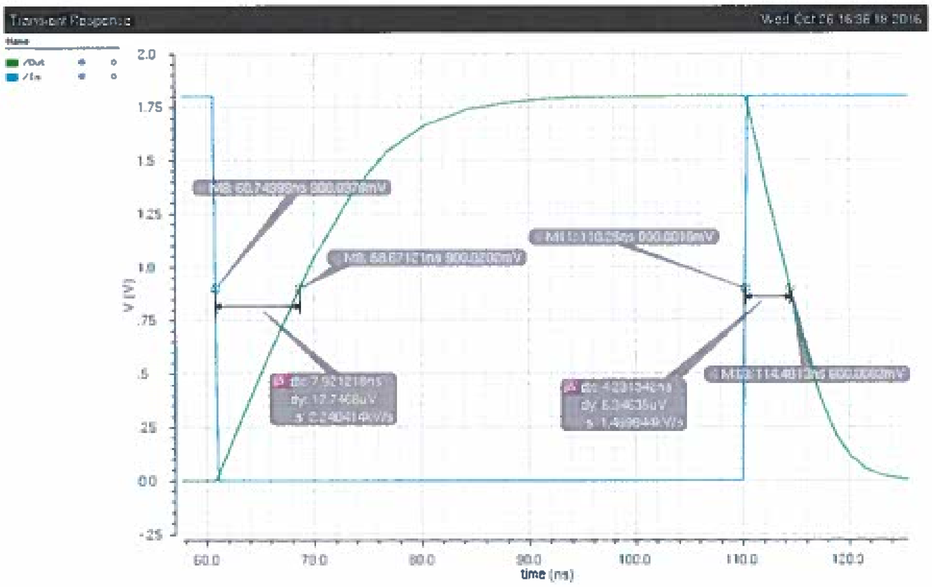


Figure 8: Propagation Delay by Fixing B and Switching A

The layout of the NAND gate can be seen in Figure 9 where the two inputs were flipped around according to the stick diagram so that in the routing of the later stage it saves some space. The DRC with LVS are illustrated in Figure 10.

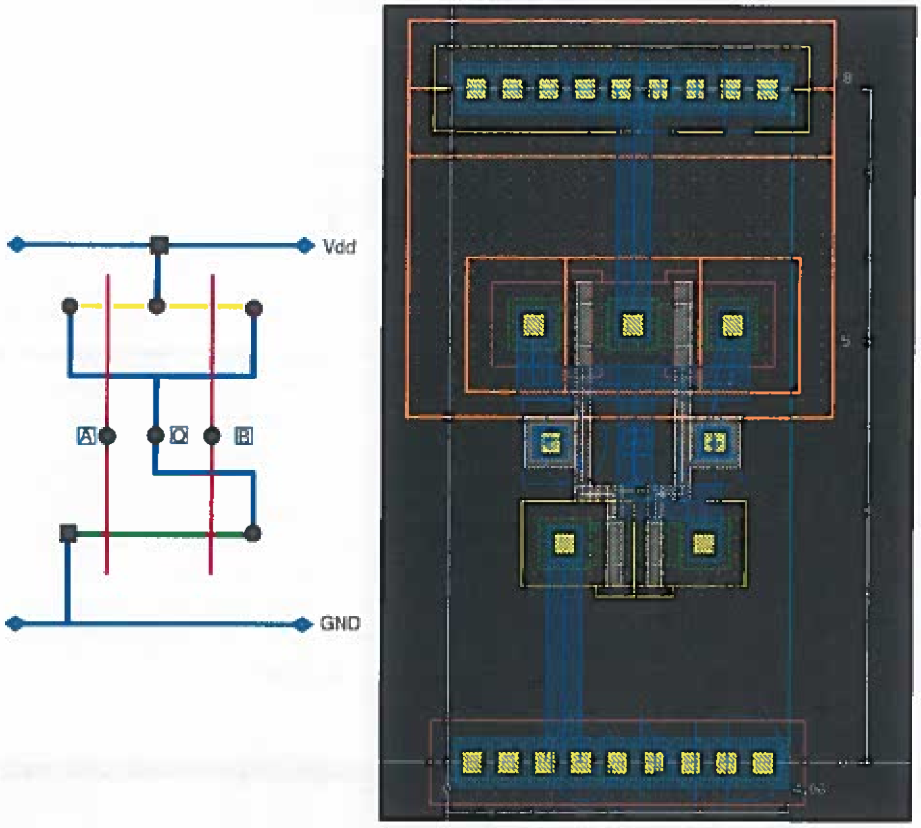
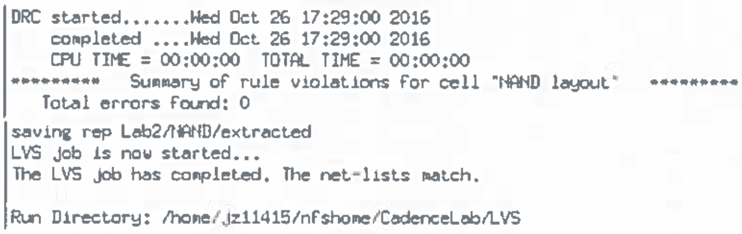


Figure 9: The Layout of the NAND Gate



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DRC started.......Wed Oct 26 17:29:00 2016
     completed ....Wed Oct 26 17:29:00 2016
     CPU TIME = 00:00:00  TOTAL TIME = 00:00:00
*********   Summary of rule violations for cell "NAND layout"   *********
    Total errors found: 0
saving rep Lab2/NAND/extracted
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/jz11415/nfshome/CadenceLab/LVS
```

Figure 10: The DRC and LVS of the NAND Gate Layout

3 D-Type Flip Flop

The design of the D-Type Flip Flop follows schematic diagram given in the lab notes where the inverter and NAND gates are the cells designed previously and the transmission gate is chosen to be $W_p/W_n = 800/400 = 2/1$. The simulation graph shows the same as it was provided in the lab notes, whereas the four vertical markers states the most important state transitions.

1. Reset = 1, D = 0, at the rising edge of the clk, the output Q follows.

2. Reset = 1, D = 1, at the rising edge of the clk, the output Q follows.

3. Reset = 0, no matter what the D or the clk is, the output Q is reset to zero.

4. Reset = 1 again, D= 0, at the rising edge of the clk, the output Q follows.

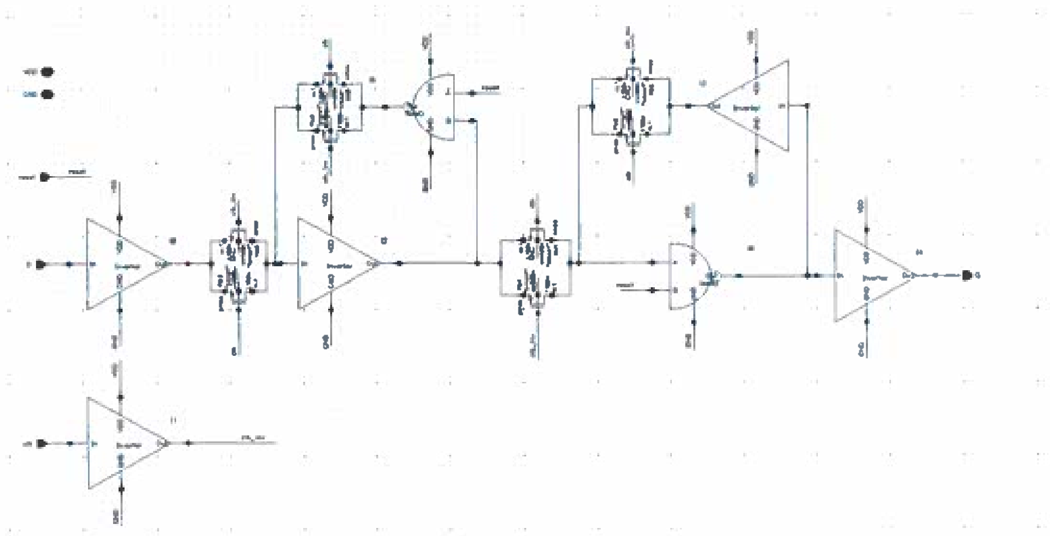


Figure 11: The Schematic of the DFF

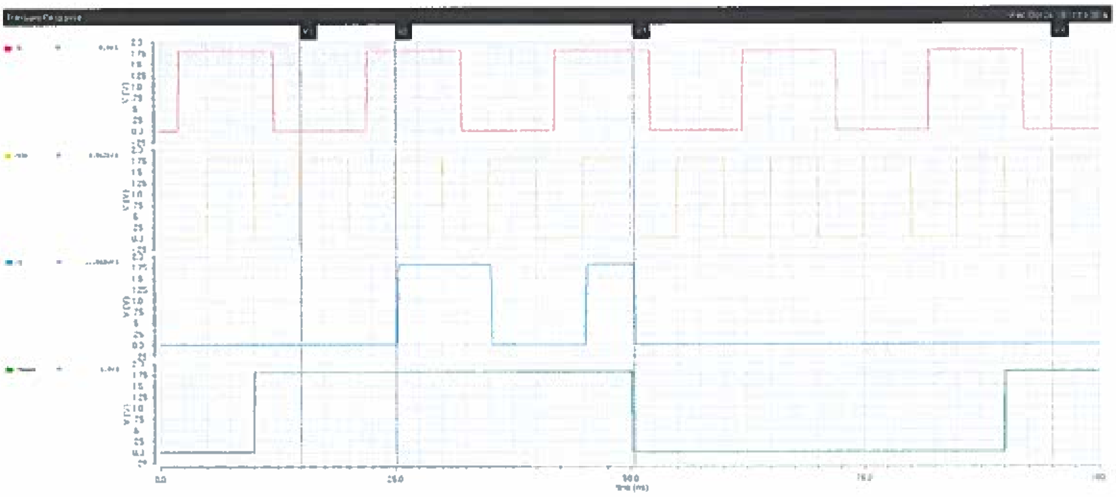


Figure 12: The Simulation of the DFF

The layout of the DFF is made to ensure it is very compact in order to minimise the space it occupied. The final size is $29.27\mu m \times 8\mu m$ where the $8\mu m$ is fixed as the standard height of this standard cell. The Pguarding are tied together to ensure all the inside part shares the same VDD and same as the Nguarding for the GND purpose.

The final layout can be seen in Figure 13 and the DRC LVS check are given in Figure 14, in which four main efforts have been done:

1. There is only one metal layer and the polysilicon used for the inner-cell routing so that the future inter-cell routing has more tolerance.
2. For each routing wire, the size of the polysilicon was minimised within the cell so that it would not provide a large resistance nor capacitance.
3. Two transmission gates are combined together by sharing the drain and the source so that it saves both the space and the connection wire in between.
4. The standard cell of inverter and NAND gates are redesigned so that they leave a large space between the PMOS and the VDD and between the NMOS and the GND. These spaces allow dual routes to go through without hitting the design rule which makes the routing a lot easier.

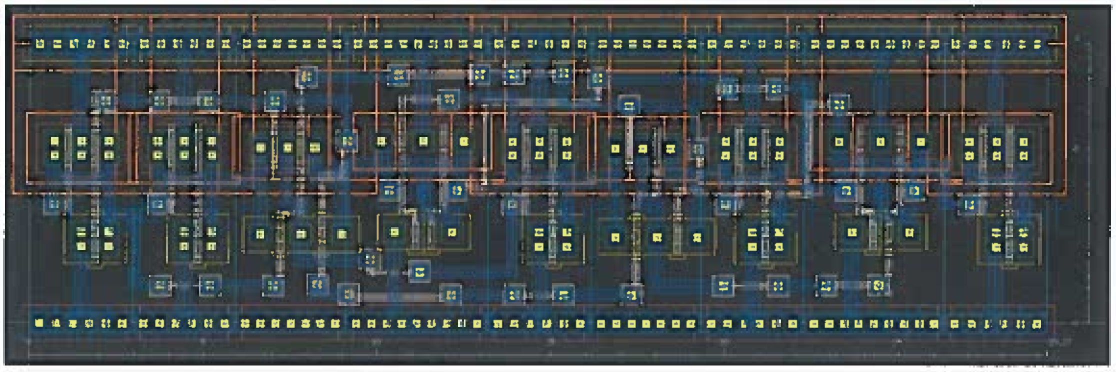
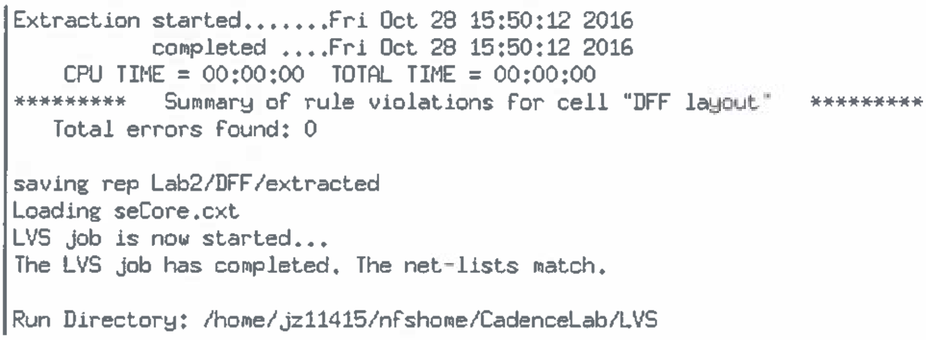


Figure 13: The Layout of the DFF



```
Extraction started......Fri Oct 28 15:50:12 2016
          completed ....Fri Oct 28 15:50:12 2016
    CPU TIME = 00:00:00  TOTAL TIME = 00:00:00
*********  Summary of rule violations for cell "DFF layout"  *********
   Total errors found: 0

saving rep Lab2/DFF/extracted
Loading seCore.cxt
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/jz11415/nfshome/CadenceLab/LVS
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Figure 14: The DRC of the DFF Layout

References

[1] R. J. Baker, *CMOS Circuit Design, Layout, and Simulation.* Wiley-IEEE Press, 3rd ed., 2010.

```
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Name: Junming Zeng (jz11415)
Assignment: Lab 2 logbook submission
Date Submitted: Friday, 28 October 2016 15:53:19 o'clock BST
Current Grade: 68.42

Submission Field:
There is no student submission text data for this assignment.

Comments:
Lab2 Logbook Submission

Junming Zeng

CID:01230747

Files:
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        Filename: Lab 2 logbook
submission_jz11415_attempt_2016-10-28-15-53-19_Lab2_Logbook_JunmingZeng.pdf
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Analogue Design Logbook for Lab3

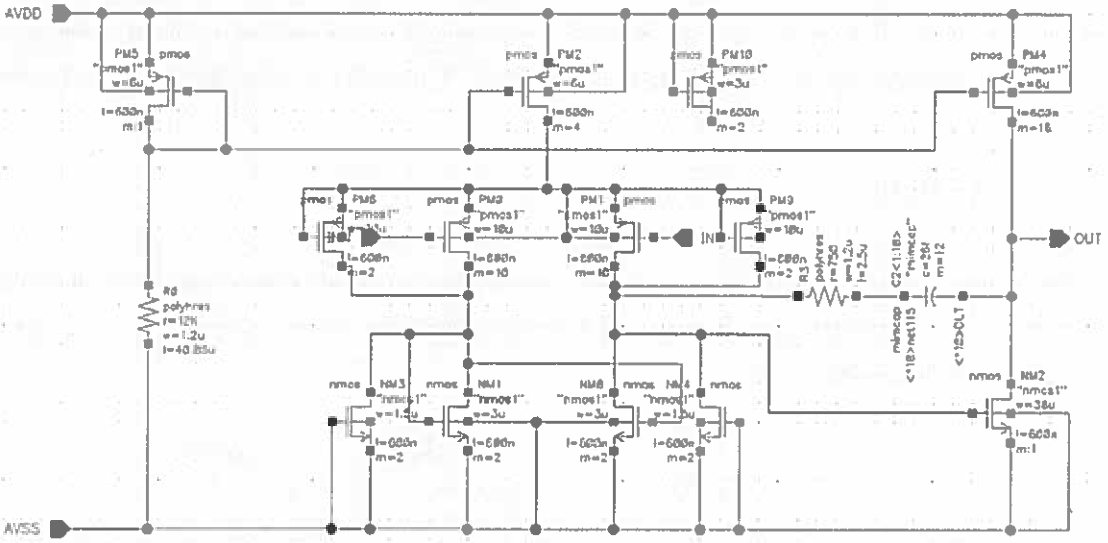
Junming Zeng

CID:01230747

November 4, 2016

1 Schematic

The schematic is shown in Figure 1.



AVDD
AVSS
OUT
IN
PM5
pmos
"pmos1"
w=6u
l=630n
m:1
PM2
m=4
PM10
w=3u
m=2
PM4
m=16
PM6
PM0
PM1
w=18u
m=16
PM3
R0
polyhres
r=12K
w=1.2u
l=40.85u
R3
r=750
l=2.5u
C2<1:16>
"mimcap"
c=25f
m=12
NM3
NM0
NM5
NM4
nmos
"nmos1"
w=1.5u
w=3u
NM2
w=38u

Figure 1: The Schematic of the Opamp

As shown in Figure 1, the top three transistors constructed a current mirror with gain where the PM2 gives 4 times the reference current defined by R0 and PM5 as the bias current for the differential pair PM0 and PM1. PM4 gives 16 times the reference current as the output current gain. The bottom NM0 and NM1 are active loads of differential pair which is also a current mirror to cancel out the common current input while amplifying the differential current input. The miller compensation capacitor C2 reduces the gain at the high frequency so that it can have a larger gain margin and phase margin.

MOS	Total Width	Length	Multipliers	Fingers
PM0	100u	600n	10	1
PM1	100u	600n	10	1
PM2	6u	600n	4	2
PM4	96u	600n	16	2
PM5	6u	600n	1	2
NM0	6u	600n	2	2
NM1	6u	600n	2	2
NM2	38u	600n	1	1

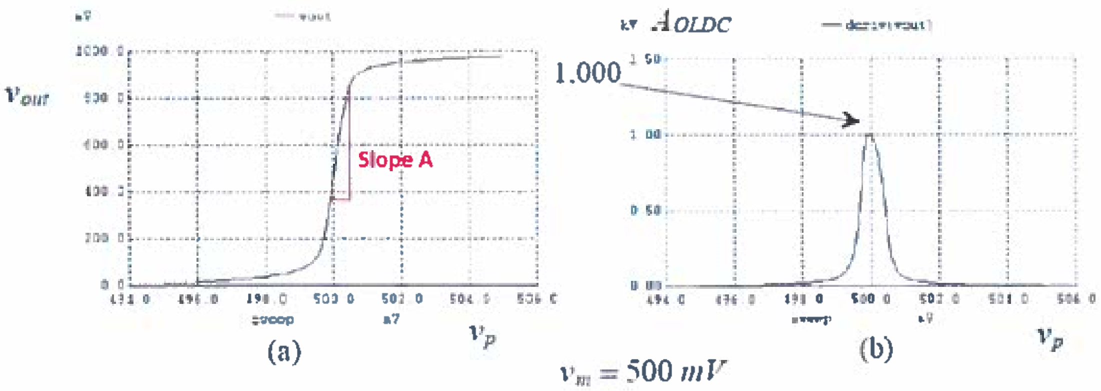
Table 1: Transistor sizes and same colour denotes the common centroid configuration

2 Calculation and Simulation

In this section, several analysis are done to estimate the impact of the OTA with respect to the operation parameter. Some simulations are applied to compare them to the theoretical result. There are three main analysis. A dc analysis to see the input linear range. An ac analysis to perform the output gain, 3-dB cut-off frequency gain margin and phase margin. A transient response to show the slew rate.

2.1 DC Analysis

This operational transconductance amplifier amplifies the input differential voltage to the output current so the g_m is given by $\frac{\partial i_o}{\partial v_d}$. By applying a constant load, the voltage gain is $g_m \times r_{load}$ and the DC response should be like Figure 2.



(a)

(b)

Figure 2: DC Transfer Curve and its Gain[1]

The input linear region is defined by region of the input when the output gradient drops by 1% of the peak value. Mathematically, [2]

$$g_m = \frac{\partial \Delta I_D}{\partial \Delta V_{in}} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\frac{\frac{4I_{bias}}{\mu_n C_{ox} W/L} - 2\Delta V_{in}^2}{\sqrt{\frac{4I_{bias}}{\mu_n C_{ox} W/L} - \Delta V_{in}^2}} \tag{1}$$

This gives $g_{mmax} = \sqrt{\mu_n C_{ox} W/L I_{bias}}$ when $\Delta V_{in} = 0$ Thus the input linear region is calculated by substitute the $g_m = 0.99 \times g_{mmax}$ (1% drop) into equation 1 and calculate the ΔVin.

2.2 AC Analysis

To analyse the gain, which is more specifically, the open loop gain, this two stage amplifier can be seen as a gain from the differential amplifier times the gain from the common source amplifier.

The gain of the first stage is given by [2]

$$A_{v1} = -\frac{g_{mn1}}{g_{on1} + g_{op1}} \tag{2}$$

where g_{mn1} is the NMOS transconductance (assuming for NMOS0 and NMOS1 are identical) and g_{on1} and g_{om1} are the output conductance in the small signal model for NMOS0/1 and PMOS0/1 respectively(assuming two PMOS are identical).

This is equivalent to

$$A_{v1} = -g_{mp1} \times (r_{on1}||r_{op1}) \tag{3}$$

Similarly, the gain for the second stage is

$$A_{v2} = -\frac{g_{mn2}}{g_{op4} + g_{on2}} = g_{mn2} \times (r_{op4}||r_{on2}) \tag{4}$$

Thus, the overall gain A is:

$$A = A_{v1} \times A_{v2} = g_{mp1} \times (r_{on1}||r_{op1}) \times g_{mn2} \times (r_{op4}||r_{on2}) \tag{5}$$

where the first stage amplifies the difference and the second stage provides current gain, and two inverse amplifier give a positive gain. Since [3]

$$g_m = \sqrt{2\mu C_{ox}\frac{W}{L}I_D} \tag{6}$$

$$r_o \approx \frac{1}{\lambda I_D} \tag{7}$$

$$\lambda \propto \frac{1}{L} \tag{8}$$

We get

$$g_m r_o = \frac{\sqrt{2\mu C_{OX}\frac{W}{L}I_D}}{\lambda I_D} \tag{9}$$

Therefore, **the change of gain with respect to W, L and I_D** is

$$A \propto g_m r_o \propto \sqrt{\frac{W/L}{I_D}} \tag{10}$$

Relate this to the overdrive voltage ($V_{GS} - VT$), recall that

$$I_D \propto (V_{GS} - VT)^2 \tag{11}$$

so that, **the gain and overdrive voltage is inverse proportional to each other** as

$$A \propto g_m r_o \propto \frac{1}{V_{GS} - V_T} \tag{12}$$

Besides, the bias current I_D is also defined as

$$I_{D2} \approx I_{ref} \times \frac{\frac{W_{p2}}{L_{p2}}}{\frac{W_{p5}}{L_{p5}}} \text{ and } I_{D4} \approx I_{ref} \times \frac{\frac{W_{p1}}{L_{p1}}}{\frac{W_{p5}}{L_{p5}}} \tag{13}$$

where the I_{ref} is defined by the power supply, the MOS overdrive voltage and the fixed resistor at the bottom.

With I_D amplifies by W/L at the output stage from the reference stage of the current mirror, the overall gain will be attenuated.

The above can be summarised in Table 2. **To increase gain within this topology, we can increase the W/L ratio for the amplifying PMOS in the differential pair and/or the W/L of the NMOS in the common source stage. Otherwise, we can also decrease the bias current based on decreasing $\frac{W_{p2}}{L_{p2}}/\frac{W_{p5}}{L_{p5}}$ and/or $\frac{W_{p1}}{L_{p1}}/\frac{W_{p5}}{L_{p5}}$ which are the two current sources from the current mirror. This is also linked to the decrease of V_{GS}, as a sense of decreasing the overdrive voltage. The cascade can also be introduced in the differential pair and the common source stage so that the output impedance is increased dramatically, and thus the overall gain can be boosted.**[4]

Effect	$V_{GS} - V_{TH}$ ↑	I_D ↑	W/L(differential pair and the common source) ↑	W/L(current mirror output stage) ↑
Gain	↓	$\sqrt{\downarrow}$	$\sqrt{\uparrow}$	$\sqrt{\downarrow}$
Cut-off frequency	↑	$\sqrt{\uparrow}$	$\sqrt{\downarrow}$	$\sqrt{\uparrow}$

Table 2: Impact on Performance According to Design Parameters

The 3dB cut-off frequency is where the first pole starts to get involve and is given by [1].

$$F_{p1} = \frac{1}{2\pi g_{m2} R_1 R_2 C_c} \tag{14}$$

where

$$R_1 = r_{on1}||r_{op1}, R_2 = r_{op4}||r_{on2}||r_{load}, g_{m2} = g_{mp4}$$

Intuitively, the term $g_m \times r$ defines the denominator of the cut off frequency expression but $g_m \times r$ is proportional to the gain. **Frequency is therefore inversely proportional to the gain and this is also a trade off between the gain and the frequency.** For example, if we want a high gain, we can do this by decreasing the biasing current I_D. However this will increase the $g_m \times r$ term ($A \propto g_m r_o \propto \sqrt{\frac{W/L}{I_D}}$), which causes the frequency to drop according to equation 14.

2.3 Transient Analysis

The transient analysis is used to get the slew rate of the op-amp and the slew rate is defined as the maximum possible rate of change of the op-amp output voltage.

$$SlewRate = \left.\frac{dV_{out}}{dt}\right|_{max} (V/\mu second) \tag{15}$$

For an OTA, the

$$v_o(t) = \frac{1}{C}\int_0^t I_{max}\cdot dt = \frac{I_{max}}{C}\cdot t \tag{16}$$

$$SlewRate = \frac{I_{max}}{C} \tag{17}$$

As seen from the equation, the slew rate is directly related to the bias current where increasing the bias current will cause an increasing in the slew rate.

2.4 Theoretical Calculation

Given the schematic diagram in Figure 1, the gain, cut-off frequency and the slew rate can be calculated to estimate the performance of the op-amp. Gain: According to equation 5 and given in static saturation, $g_{mp1} = 1.32233m$, $r_{op1} = 111.036K$, $r_{on1} = 139.741K$, $g_{mn2} = 5.9619m$, $r_{op4} = 8.66015K$ and $r_{on2} = 29.1744K$.

$$\begin{aligned} A &= A_{v1}\times A_{v2} \\ &= g_{mp1}\times(r_{on1}||r_{op1})\times g_{mp4}\times(r_{op4}||r_{on2}) \\ &= 1.32233m\times\frac{139.741K\times 111.036K}{139.741K+111.036K}\times 5.9619m\times\frac{8.66015K\times 29.1714K}{8.66015K+29.1714K} \\ &= 3257 = 70dB \end{aligned} \tag{18}$$

Cut-off Frequency: Based on the equation 14, and the same parameter as used in calculating the gain.

$$F_{3dB} = \frac{1}{2\pi g_{mp4}(r_{op1}||r_{on1})(r_{op4}||r_{on2}||r_{load})C_c} = 10.74KHz \tag{19}$$

Slew rate: The output switched in a given speed dependent on the bias current of the common source stage charging and discharging the capacitor. Given $I_d = -1.08254m$ and $C_c = 3pF$

$$SlewRate = \frac{I_D}{C} = \frac{-231.8m}{3\times 10^{-9}} = 77.27(MV/S) \tag{20}$$

2.5 Simulation

In the simulation part, the general response of the op-amp will be shown. Then, by sweeping the parameters including biasing current and the W/L, the graphs can justify the calculations. The first simulation is based on the designed parameter and give out the desired response. The gain and frequency response is given in Figure 3.

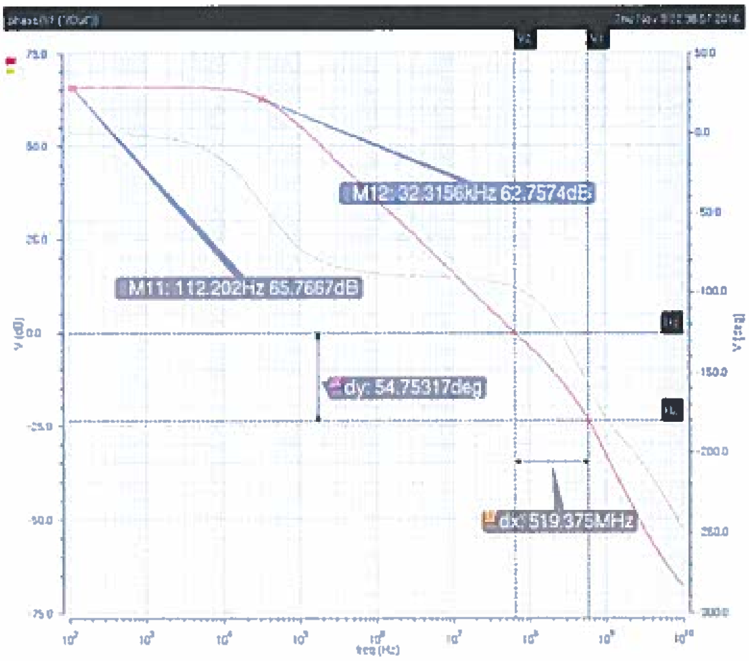


Figure 3: The Gain and Phase of the Opamp

This gives a 65.76dB DC gain and a 32.3KHz bandwidth. The gain margin and phase margin are also stated on the graph. Compare this with the calculation in the previous section. The gain is 4dB smaller and the bandwidth is 3 times larger.

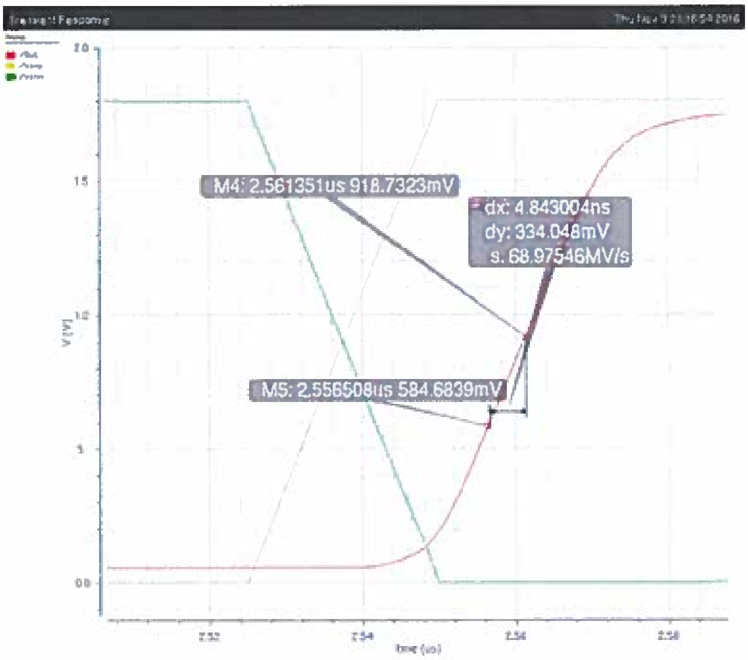


Figure 4: The Transient Response of the Opamp

Figure 4 shows the transient response while the differential input switch from -1.8V to +1.8V. The gradient of the output switching curve defines the slew rate and it is given by 69 (MV/S) which is similar to the theoretical calculation.

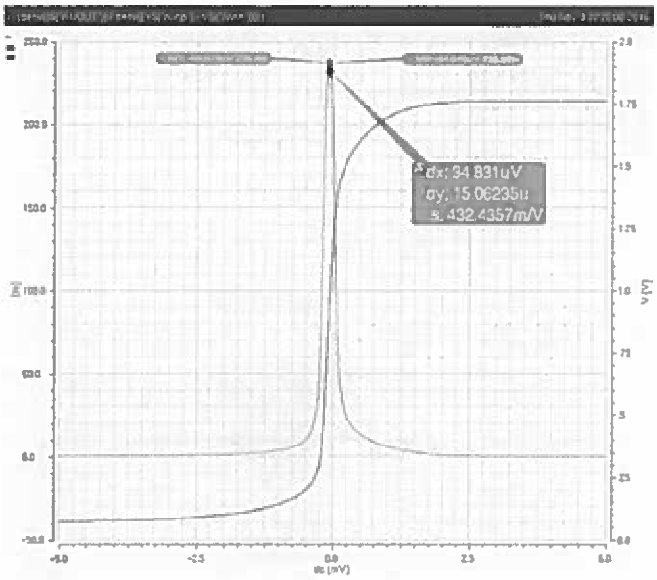


Figure 5: The Input Linear Range of the Opamp

Figure 5 shows that the input linear range is measured by the derivative of the output current compared to the input differential voltage(in this case modulated to 1). By taking the 1% off of the peak value, the effective linear range of this design is 34.83 uV.

The following graphs will contain the gain of the op-amp by sweeping the design parameters.

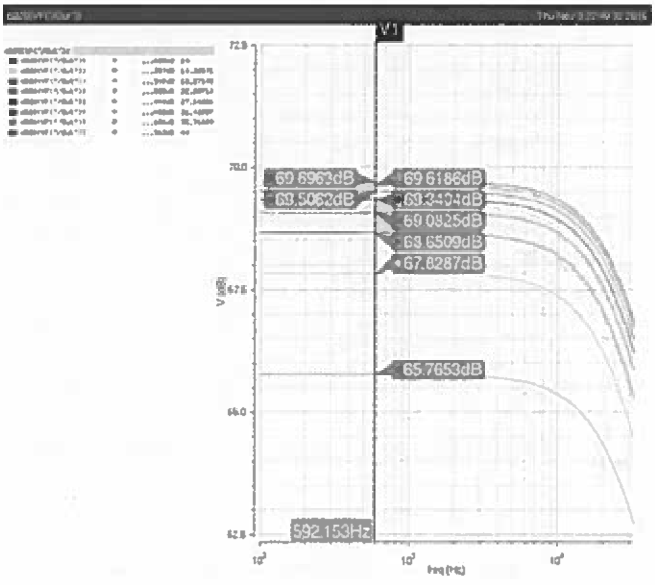


Figure 6: The Gain of the Opamp with respect to W/L at the gain stage

Z

Figure 6 shows by increasing the W/L at the gain stage which includes the differential pair PMOS and the common source NNOS, the overall gain increases nonlinearly, as $A \propto g_m r_o \propto \sqrt{\frac{W/L}{I_D}}$.

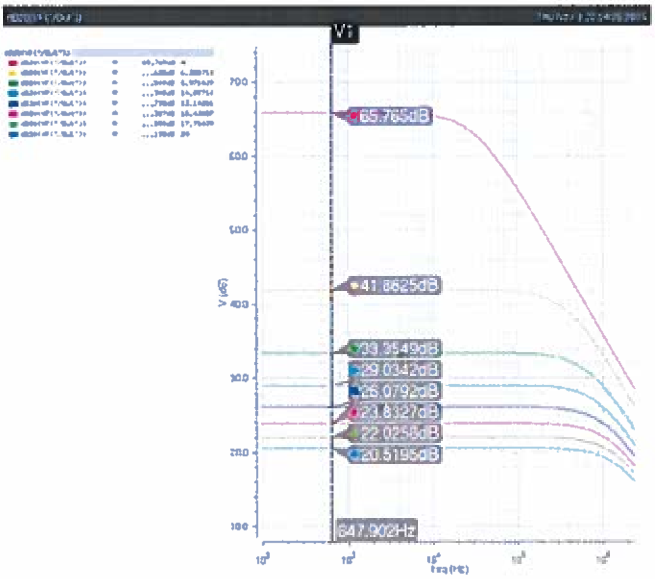


Figure 7: The Gain of the Opamp with respect to W/L at the current mirror output stage

Figure 7 shows that by increasing the W/L at the current mirror output stage, the effective increase in biasing current will cause the gain to be fallen which also follow that $A \propto g_m r_o \propto \sqrt{\frac{W/L}{I_{bias}}}$

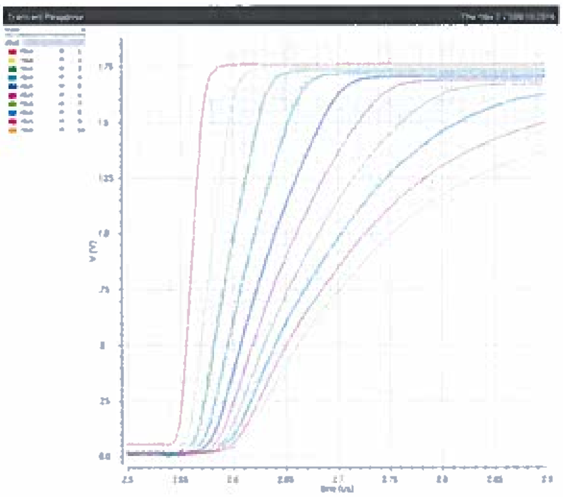


Figure 8: The Slew Rate by increasing the biasing current

Figure 8 shows that by increasing the biasing current, the transaction curve gets sharper and the relative gradient increases, which is proved by $SlewRate = \frac{I_D}{C}$.

Generally, the gain and the frequency are trade-off pairs. The relationship of the design parameters and those two are just the opposite. Due to the lack of space, the frequency response are not listed but they match as the theory as well.

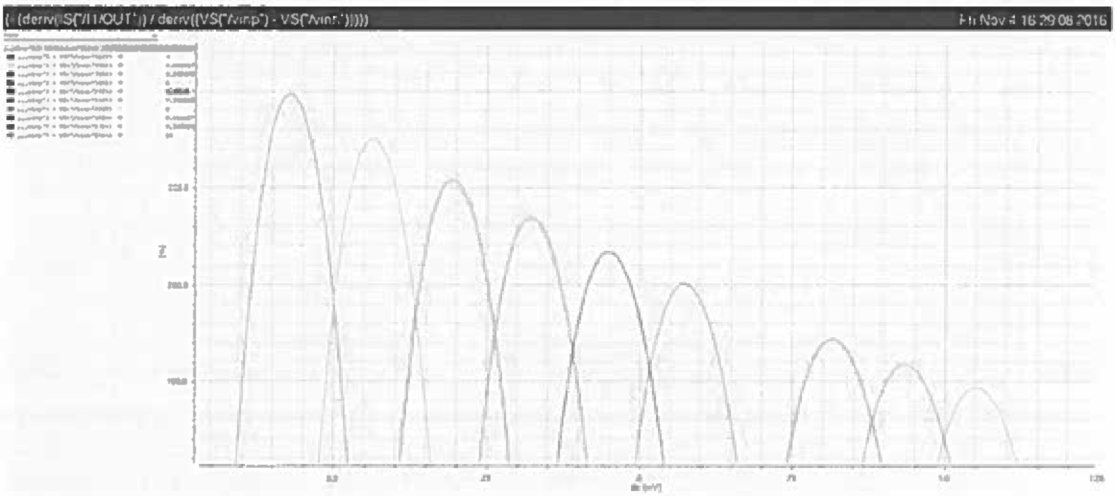


Figure 9: The Input Linear Range by increasing the W/L

Figure 9 shows how the input linear range varies by sweeping the bias current, as the bias current increases, the input linear region gets larger and larger.

3 Layout

The layout of this OTA is shown in Figure 10 and the DRC LVS check are followed. The transistor sizes are shown in Table 1 and several considerations have been done.

1. All the current mirrors and differential pair are made common centroid because they are sensitive. The arrangement used in this case are CCCCCCCCBBABBCCCCCCCC and

   DBAABBAABBAD

   DABBAABBAABD

   where A B C represent the common centroid transistors and D represents the dummy transistor.
2. Dummy transistors are added at every edge of the active area so that the source and drain are better aligned to reduce the edge effect.
3. Guard rings are added to provide isolations between small blocks.
4. The capacitor has been divided to 120 small capacitors so that it is better for a compact layout.
5. The common source NMOS is not divided to multipliers because it will cause the simulation to show that the overall gain decreases seriously.
6. The arrangement between blocks has been optimised with an overall size of $91.45um \times 48.65um$.

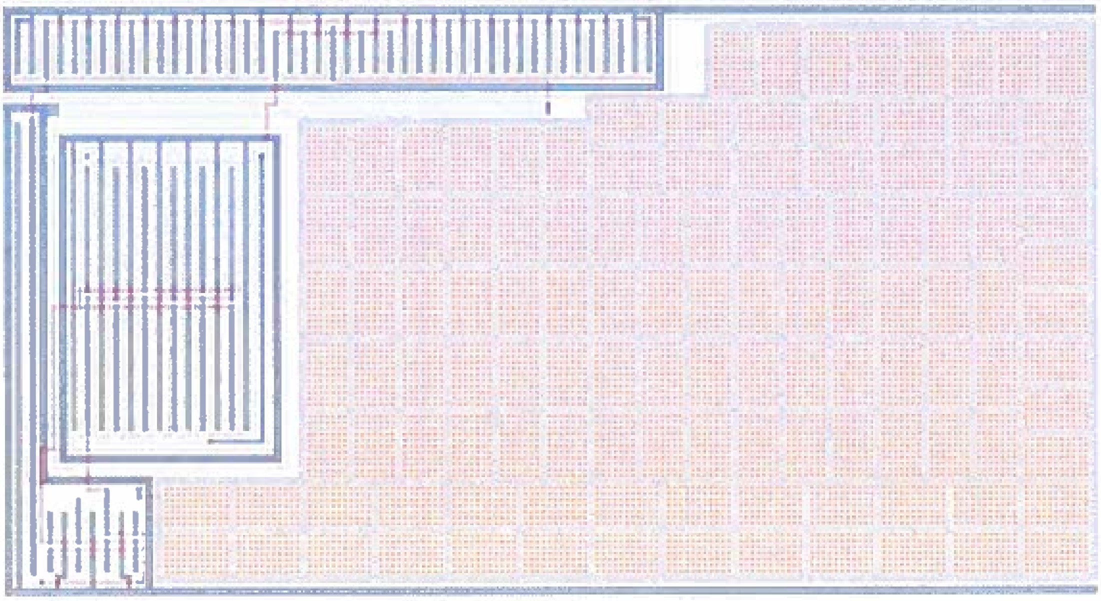
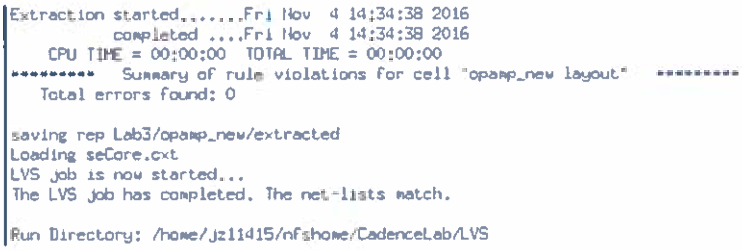


Figure 10: The Layout of the opamp



```
Extraction started.......Fri Nov  4 14:34:38 2016
             completed ....Fri Nov  4 14:34:38 2016
     CPU TIME = 00:00:00  TOTAL TIME = 00:00:00
*********   Summary of rule violations for cell "opamp_new layout"   *********
    Total errors found: 0

saving rep Lab3/opamp_new/extracted
Loading seCore.cxt
LVS job is now started...
The LVS job has completed. The net-lists match.

Run Directory: /home/jz11415/nfshome/CadenceLab/LVS
```

Figure 11: The DRC LVS of the opamp

References

[1] R. J. Baker, *CMOS Circuit Design, Layout, and Simulation*. Wiley-IEEE Press, 3rd ed., 2010.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*. New York, NY, USA: McGraw-Hill, Inc., 1 ed., 2001.

[3] A. Sedra and K. Smith, *Microelectronic Circuits*. No. v. 1 in Microelectronic Circuits, Oxford University Press, 1998.

[4] P. R. Gray, B. A. Wooley, and R. W. Brodersen, *Circuit Concepts and Design Techniques*, pp. 3–157. Wiley-IEEE Press, 1989.

```
          Lab 3 logbook submission_jz11415_attempt_2016-11-04-16-43-09
Name: Junming Zeng (jz11415)
Assignment: Lab 3 logbook submission
Date Submitted: Friday, 4 November 2016 16:43:09 o'clock GMT
Current Grade: 77.36

Submission Field:
There is no student submission text data for this assignment.

Comments:
Lab 3 logbook submission

Junming Zeng

01230747

Files:
        Original filename: Lab3.pdf
        Filename: Lab 3 logbook 
submission_jz11415_attempt_2016-11-04-16-43-09_Lab3.pdf
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